Varvara Nazarenko

AI/ML Engineer · Solutions Architecture

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Experience

Diamond mining company Jul 2025 – Present

ML / Data Analyst — Pricing & Market Intelligence

Company name available on request

- Worked in a small technical team directly with the Head of Pricing and business stakeholders, translating open-ended pricing questions into data products and analytical solutions built from scratch.
- Built analysis-ready datasets from ∼30 GB of multi-source market data using Python and PostgreSQL, including data cleaning, aggregation and automated reporting workflows.
- Designed weighted price indices and a liquidity matrix that reduce multidimensional market data into interpretable metrics for product and pricing comparisons across B2B and B2C markets.
- Applied statistical and classical ML methods to pricing, segmentation, anomaly detection and product selection; delivered internal Python tools, automated reports and BI dashboards.

HSE University Aug 2025 – Present

Research Assistant — Machine Learning, Faculty of Computer Science

Top-ranked Russian research university

- Developed PIEFS, a PyTorch representation-learning method based on spectral and geometric ML, covering method design, implementation and experimental validation.
- Built the research codebase from scratch as a modular experimental framework with configurable training, reproducible evaluation and Weights & Biases tracking.
- Designed experiments and diagnostic controls to test learned representations, including analyses that exposed failure modes of the learned metric.
- Worked closely with the research supervisor through weekly technical reviews, turning experimental findings into an ICML 2026 workshop paper, technical reports and the B.Sc. thesis.

Institute of Bioorganic Chemistry, Russian Academy of Sciences Nov 2024 – May 2025

Undergraduate Research Project — Computational Biology

- Prepared TCGA datasets from 10 cancer cohorts (∼59K mRNA features) and built models to predict ∼1.9K miRNA targets.
- Compared feature-selection methods and modern tabular models, including GANDALF; reached Spearman $\rho \approx 0.8$ on selected targets and found that pooling cohorts worked better than separate cancer-specific models.

Selected Publication

PIEFS: Physics-Informed Eigenfunction Features with Learnable Scaling Jul 2026

First Author · AI4Physics Workshop, ICML 2026 · arXiv

Selected Project

Lumni — LLM System for Marketing-Brief Generation Dec 2025

Vertex AI · Supabase · Embeddings

- Designed a brand-DNA representation and retrieval layer that structures brand information for similarity search.
- Built Vertex AI workflows for brand enrichment and marketing-brief generation and connected them to Supabase.

Education

Higher School of Economics (HSE) Sep 2022 – Jul 2026

B.Sc. Applied Mathematics & Computer Science, Machine Learning track Moscow, Russia

- **Final-semester GPA**: 8.56/10, top 5% of the cohort; B.Sc. thesis on PIEFS
- **Relevant Coursework**: Machine Learning, Deep Learning, MLOps & ML Systems, Applied Statistics, Optimization

Additional Education

Moscow School of Programmers — Yandex campus: 450-hour program in C#, computer networks and CTF security

Lyceum No. 40, Nizhny Novgorod: Physics track affiliated with IAP RAS; top-50 Russian STEM school

Skills

Programming: Python, SQL, C++

ML & Data: PyTorch, scikit-learn, Hugging Face, NumPy, pandas, Polars

Tools & Platforms: PostgreSQL, Docker, Git, Hydra, Vertex AI, Supabase, Weights & Biases, Power BI

Languages: English (C1), Russian (native), Hebrew (basic), French (basic), Italian (basic)